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Performance Shortfalls in Saudi EPC Projects: Testing, Output Guarantees, and Remedies under the Civil Transactions Law

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Abstract

In engineering, procurement and construction (EPC) projects, physical completion is not necessarily contractual completion. A power plant may be mechanically complete yet fail to deliver the contracted megawatts; a processing facility may operate while consuming more feedstock or utilities than promised; and a renewable-energy asset may export electricity while falling below its guaranteed performance ratio. In each case, the employer has received an asset, but not quite the asset for which the contractual price was negotiated. This paper examines how Saudi law should address that problem following the enactment of the Civil Transactions Law (CTL). Its central argument is that the CTL is unusually compatible with the performance-based logic of EPC contracting. Article 168 expressly distinguishes an obligation to exercise care from an obligation to achieve a specified result, and treats the latter as performed only when the promised result is achieved. Properly drafted output guarantees therefore have a statutory footing that goes beyond ordinary workmanship obligations. Yet the remedial consequences of a shortfall are not mechanical. The CTL also regulates cure and termination, causation, contributory fault, contractual compensation, and judicial moderation of pre-agreed damages. Particular attention is given to Article 179, which permits reduction of agreed compensation where the obligation has been partially performed. The paper argues that Saudi EPC contracts should respond by separating minimum or absolute performance requirements—below which the employer may reject the works or refuse acceptance—from economic performance guarantees, for which a calibrated performance-damages formula prices an agreed degree of underperformance. Public Saudi project disclosures, FIDIC practice, and comparative engineering cases demonstrate that this two-tier architecture is commercially established. Under the CTL, it is also legally coherent. The resulting framework can improve drafting, testing discipline, dispute avoidance, and the defensibility of performance remedies in Saudi energy and industrial projects.

Introduction

Saudi Arabia's current project market is defined not only by the volume of construction, but by the increasing technical complexity of the assets being procured. Power generation, water treatment, petrochemicals, mining, renewables, data infrastructure and process facilities are not purchased merely as assembled concrete, steel and equipment. They are purchased for what they will do. The economic premise of an EPC bargain is therefore functional: the contractor undertakes to deliver an integrated facility capable of producing a defined output, at a defined quality, within defined consumption, reliability, emissions or efficiency limits. The enactment of the Civil Transactions Law gives this familiar commercial proposition a newly codified legal framework in Saudi Arabia.¹ The CTL entered into force in December 2023 and now supplies both general rules of contractual obligation and a specific statutory regime for contracts for service, including construction or muqawala contracts.²

This matters because a performance shortfall is conceptually different from an ordinary defect. A cracked wall, a leaking flange or an incorrectly installed cable can generally be identified as non-conforming work and repaired. A plant that is fully constructed but achieves only 96 percent of guaranteed output presents a more difficult question. The facility may be capable of commercial operation; the physical works may be substantially complete; and the employer may prefer to operate rather than reject the project. Yet four percent of permanent output can represent a material loss over a twenty- or thirty-year operating life. The question is therefore not simply whether the contractor breached the contract. It is what the law should make of a delivered asset whose physical existence is complete but whose economic performance is not.

International EPC forms have long addressed this problem through a sequence of testing, taking over, correction, retesting, performance guarantees and pre-agreed damages. The FIDIC Silver Book is the most prominent example. Its 2017 edition is expressly designed for EPC/turnkey delivery and retains a detailed contractual architecture for Tests on Completion, taking over, defects, Tests after Completion and remedies.³ FIDIC's own description of the suite makes the point plainly: under the Silver Book, taking over is linked to successful completion of the relevant tests.⁴ The form does not treat testing as an administrative afterthought. It uses testing to

¹ Civil Transactions Law, Royal Decree No. M/191 (18 June 2023), Bureau of Experts at the Council of Ministers, Official Translation Department, <https://misa.gov.sa/app/uploads/2025/07/Civil-Transactions-Law.pdf> (official English translation; Arabic text governs).

² Nabeel Ikram & Giuliano Parlascino, Saudi Arabia's Civil Transactions Law, INTERNATIONAL BAR ASSOCIATION (25 July 2024), noting that the CTL entered into force on 16 December 2023, <https://www.ibanet.org/clint-june-2024-feature-3>.

³ Fédération Internationale des Ingénieurs-Conseils (FIDIC), CONDITIONS OF CONTRACT FOR EPC/TURNKEY PROJECTS (Silver Book), 2d ed. 2017, reprinted 2022 with amendments, <https://fidic.org/books/epcturnkey-contract-2nd-ed-2017-silver-book-reprinted-2022-amendments>.

⁴ FIDIC, THE FIDIC SUITE OF CONTRACTS, explaining that under the Silver Book taking over follows successful completion of the contractual tests, https://fidic.org/sites/default/files/FIDIC_Suite_of_Contracts_0.pdf.

determine whether the contractor has delivered the performance that the price and risk transfer were intended to purchase. This is not merely an international reference point: a 2026 Saudi construction-law survey identifies FIDIC forms as the most commonly used construction contracts in the Kingdom and notes that they are frequently modified, often through extensive employer-favouring risk reallocations.⁵

The same commercial logic appears outside standard forms. Public disclosures concerning Saudi industrial projects show contracts in which provisional acceptance depends on performance tests, minimum performance thresholds operate as a floor, and liquidated damages compensate for a measurable shortfall above that floor.⁶ Ma'aden has likewise disclosed EPC arrangements under which failure to achieve guaranteed performance can trigger performance liquidated damages in Saudi-governed contracts.⁷ These disclosures are useful because they show that the distinction between an asset that is unacceptable and an asset that is acceptable at a discounted performance level is not theoretical. It is embedded in actual project finance and industrial contracting practice in the Kingdom.

The CTL now requires that this contractual machinery be examined through Saudi statutory concepts. Article 168 is the starting point. It distinguishes between an obligation satisfied by reasonable care and an obligation to achieve a specific result; in the latter case, performance occurs only when the result is achieved.⁸ That rule is exceptionally important for EPC drafting. It provides a principled basis for treating an output guarantee not merely as evidence of quality, but as an obligation of result. If the contractor promises a defined net output, heat rate, throughput, availability or emissions limit, and the contract makes that metric a required result, it is difficult to characterize delivery of something materially different as complete performance merely because the plant has been built and can operate.

Article 168 does not, however, answer the remedial question by itself. The CTL also gives the client rights to require correction, engage another contractor in appropriate circumstances and seek termination; it permits agreed compensation; it authorizes reduction of agreed compensation where the obligation has been partly performed; and it reduces compensation where the

⁵ Alex Saleh, Hegui Taha, May El Mahdy & Omar Halbouni, Saudi Arabia, in CONSTRUCTION LAW 2026, CHAMBERS GLOBAL PRACTICE GUIDES (updated 4 June 2026), noting that FIDIC forms remain the most common construction contracts in Saudi Arabia and are frequently amended, <https://practiceguides.chambers.com/practice-guides/construction-law-2026/saudi-arabia>.

⁶ Advanced Polypropylene Company, Prospectus 47 (16 Nov. 2006), Capital Market Authority, describing provisional acceptance, minimum performance levels and liquidated damages for performance shortfalls, <https://cma.gov.sa/en/Market/Prospectuses/Documents/APPC.pdf>.

⁷ Saudi Arabian Mining Company (Ma'aden), Rights Issue Prospectus 213, 221, 227 (2014), Capital Market Authority, describing performance liquidated damages under Saudi-governed EPC contracts, <https://cma.gov.sa/en/Market/Prospectuses/Documents/Rights%20Issue%20Prospectus-Saudi%20Arabian%20Mining%20Company.pdf>.

⁸ Civil Transactions Law, Royal Decree No. M/191, art. 168.

creditor's own conduct contributes to the harm.⁹¹⁰¹¹ The interaction of these provisions means that Saudi law supports strong performance obligations while resisting remedial over-simplification.

This paper advances a two-tier model for Saudi EPC contracts. The first tier consists of minimum or absolute performance requirements. These are performance floors below which the employer has not received a commercially usable version of the promised asset and should ordinarily retain rights of cure, retesting, rejection or termination. The second tier consists of economic performance guarantees. These are targets above the minimum floor where a shortfall can rationally be priced by a formula reflecting the economic burden of lower output or higher consumption. Comparative case law already recognizes the distinction between absolute and economic performance guarantees.¹² Saudi project disclosures demonstrate a similar architecture.¹³ The paper argues that this model fits the CTL particularly well because it reconciles Article 168's result-based conception of performance with Article 179's mandatory control of pre-agreed compensation.

The analysis proceeds in eight parts. It first explains why performance is the economic product in an EPC transaction and why mechanical completion should not be confused with functional delivery. It then considers Article 168 and the related CTL provisions governing specification, intended purpose, interpretation and good faith. The paper next examines cure, retesting, taking over and rejection before turning to performance liquidated damages and Article 179. It then addresses the relationship between performance damages and other remedies, testing integrity and employer-caused interference. A comparative section draws lessons from English engineering decisions, not as sources of Saudi law but as practical demonstrations of recurring contractual problems. The final section translates the analysis into drafting and project-counsel recommendations for Saudi EPC projects.

Performance as the Product in EPC Delivery

Physical Completion and Functional Delivery

A conventional description of construction focuses on the production of a physical thing. EPC contracting is different because the 'thing' is often inseparable from its operating characteristics. A combined-cycle power plant is not commercially equivalent to another plant of identical dimensions that produces less electricity, consumes more fuel, or cannot maintain its promised

9 Civil Transactions Law, art. 466.

10 Civil Transactions Law, art. 179(2).

11 Id. arts. 128, 172.

12 *Energy Works (Hull) Ltd v. MW High Tech Projects UK Ltd* [2022] EWHC 3275 (TCC), paras. 63–65 (distinguishing absolute and economic performance guarantees in an EPC energy project).

13 *Advanced Polypropylene Company*, Prospectus 47 (16 Nov. 2006), Capital Market Authority, describing provisional acceptance, minimum performance levels and liquidated damages for performance shortfalls, <https://cma.gov.sa/en/Market/Prospectuses/Documents/APPC.pdf>.

availability. In a process plant, throughput, feedstock consumption and product quality may matter more to project economics than the visual completeness of the works. The contract therefore converts engineering assumptions into legal promises through Employer's Requirements, technical schedules, performance guarantees and test procedures.

The FIDIC literature reflects this functional conception. Detailed works requirements are typically stated through the Employer's Requirements, while the testing and taking-over regime establishes when the contractor's design and construction have translated into a functioning asset.¹⁴ Specialist commentary on the Silver Book similarly emphasizes that the form is designed for substantial contractor responsibility and a high degree of certainty as to the end product.¹⁵ Baker and his co-authors treat product obligations, including fitness, quality and testing, as central rather than peripheral features of the FIDIC bargain.¹⁶

The distinction is particularly visible in plant contracts. In *Matthew Hall Ortech Ltd v. Tarmac Roadstone Ltd*, the court described a contractual sequence running from construction and start-up, through taking over, to performance tests and acceptance. It observed that performance guarantees may address capacity, raw-material or utility consumption, product quality, reliability and similar criteria, and that reduced performance may be accepted with an associated liquidated-damages consequence.¹⁷ Although the case arose under English law and an older process-plant form, its commercial anatomy remains recognizable in modern EPC transactions: construction creates the plant; testing proves what the plant can do; acceptance decides whether the proved performance is contractually sufficient.

That anatomy matters because the word 'completion' is often used imprecisely. Mechanical completion may mean that equipment is installed and construction checks are substantially finished. Ready-for-start-up status may mean that energization or feed introduction can safely occur. Taking over may transfer care, custody or operational responsibility. Provisional acceptance may depend on a performance test. Final acceptance may follow a reliability period or defects regime. Unless the contract carefully distinguishes these events, a dispute over performance can quickly become a dispute over whether the employer has already accepted the asset, whether liquidated damages continue to run, whether a performance test remains possible, and whether continued operation amounts to waiver.

¹⁴ William Godwin QC, *THE 2017 FIDIC CONTRACTS: THE SECOND EDITIONS OF THE RED, YELLOW AND SILVER BOOKS* (Wiley 2020), doi:10.1002/9781119514619.

¹⁵ Jakob B. Sørensen, *FIDIC SILVER BOOK: A COMPANION TO THE 2017 EPC/TURNKEY CONTRACT* (ICE Publishing 2019), doi:10.1680/fsb.64362.

¹⁶ Ellis Baker, Ben Mellors, Scott Chalmers & Anthony Lavers, *FIDIC CONTRACTS: LAW AND PRACTICE*, chs. 3 & 8 (Informa Law from Routledge 2009).

¹⁷ *Matthew Hall Ortech Ltd v. Tarmac Roadstone Ltd* [1997] EWHC Technology 352, paras. 19–21, 47, 50–51 (discussing plant testing, performance guarantees, acceptance and reduced performance coupled with liquidated damages).

Minimum Performance and Economic Performance

The most useful analytical distinction is between minimum performance and economic performance. Minimum performance identifies the point below which the asset is not an acceptable substitute for the promised facility. A power plant that cannot safely sustain a defined minimum load, a desalination plant that cannot produce water meeting mandatory quality standards, or a processing facility that cannot meet an essential throughput floor may fail at the level of commercial identity. The remedy cannot sensibly be limited to a price adjustment if the employer has purchased an asset that cannot perform the function for which the project exists.

Economic performance guarantees address a different problem. Once the asset clears the minimum threshold, a smaller shortfall may be capable of valuation. A plant that produces 1,470 MW rather than 1,500 MW may remain commercially usable, but the missing 30 MW has an economic cost. A heat-rate shortfall may increase fuel expenditure over the operating life. Excess auxiliary consumption may reduce net saleable output. In that range, an agreed formula can operate as a price for underperformance while avoiding the destructive consequence of rejecting a functioning facility.

This two-level structure is visible in Energy Works (Hull), where the EPC contract differentiated ‘Absolute Performance Guarantees’ from ‘Economic Performance Guarantees.’ Failure of an absolute guarantee could entitle the employer to reject the plant, whereas economic guarantees were associated with measured performance consequences.¹⁸ The terminology is not universal, but the underlying distinction is sound and should be deliberately adopted in Saudi EPC drafting.

The Advanced Polypropylene Company prospectus supplies a Saudi-market example of the same logic. Its summary of the EPC structure states that provisional acceptance could occur either where the plant met the performance guarantees or, where applicable, where it met minimum performance levels and the contractor paid liquidated damages for the shortfall. It also states that liquidated damages were not sufficient to discharge the obligation to satisfy the minimum levels.¹⁹ This is a commercially sophisticated allocation: a specified degree of underperformance is monetized, but failure below the viability floor is not purchased by damages.

Such structures also explain why performance damages should not be treated as ordinary delay damages with a different label. Delay damages compensate for a temporal failure. Performance damages compensate for a permanent or long-lived degradation in the asset’s economics. The formula therefore needs to be linked to the relevant variable—lost output, increased consumption, reduced recovery, increased emissions-control cost or another measurable

¹⁸ Energy Works (Hull) Ltd v. MW High Tech Projects UK Ltd [2022] EWHC 3275 (TCC), paras. 63–65 (distinguishing absolute and economic performance guarantees in an EPC energy project).

¹⁹ Advanced Polypropylene Company, Prospectus 47 (16 Nov. 2006), Capital Market Authority (performance guarantees, minimum performance levels and provisional acceptance).

consequence—rather than simply imported from the daily rate used for delay. The legal treatment of performance damages under Saudi law should follow that functional difference.

Performance Guarantees as Obligations of Result under Saudi Law

Article 168 and the Legal Character of the Promise

The CTL's most important contribution to this subject appears in Article 168. Where an obligation is one of care, reasonable care may constitute performance even if the intended purpose is not achieved. Where a specific result is required, however, the obligation is performed only if the result is achieved.²⁰ The distinction gives Saudi law a vocabulary that maps closely onto modern EPC contracting. An EPC contractor may owe both types of obligation within the same project: professional care in some activities, but a strict result in others.

The classification must depend on the contract. Not every engineering statement should become an absolute warranty. A design criterion may identify an assumption, a target or an input into design calculations rather than an independently guaranteed outcome. Conversely, a schedule expressly titled 'Performance Guarantees,' accompanied by a defined test, correction curves, tolerances and a specific remedial consequence, is powerful evidence that the parties intended a result obligation. Article 104 reinforces this contextual approach by requiring the contract to be interpreted as a whole and, where interpretation is needed, by reference to mutual intent, custom, the circumstances of the transaction and consistency among the contractual provisions.²¹

The importance of precise drafting is illustrated comparatively by *MT Højgaard v. E.ON*. There the contractor had complied with an industry standard that contained a technical error, but the contract also contained a more demanding requirement concerning design life. The United Kingdom Supreme Court treated the contractual result requirement as effective notwithstanding compliance with the prescribed standard.²² The case is not authority on Saudi law, but it illustrates a recurring EPC danger: compliance with a method or standard does not necessarily discharge a separately promised outcome. Article 168 makes that conceptual distinction particularly relevant in Saudi-governed projects.

A well-drafted Saudi EPC contract should therefore identify which technical statements are minimum inputs, which are design criteria, which are targets, and which are guaranteed results. Treating every number in an Employer's Requirements as an absolute guarantee would create unmanageable risk and invite inconsistent remedies. Treating none of them as guaranteed results would deprive performance testing of much of its legal purpose. Article 168 permits a more

²⁰ Civil Transactions Law, Royal Decree No. M/191, art. 168.

²¹ *Id.* art. 104.

²² *MT Højgaard A/S v. E.ON Climate & Renewables UK Robin Rigg East Ltd* [2017] UKSC 59, paras. 25–34, 51–54.

disciplined approach: the contract can deliberately identify the results for which outcome responsibility is transferred to the contractor.

Specification, Intended Purpose, and the Contract as a Whole

The specific muqawala provisions support that analysis. Article 461 defines the contract for service as one in which the contractor makes a thing or carries out work for a fee without subordination or agency. Article 463 requires contractor-supplied materials to satisfy agreed requirements and specifications; where no specific requirements are agreed, the materials must be sufficient for the intended purpose according to custom.²³ Article 465 further requires the contractor to complete the work in accordance with the terms of the contract and within the agreed period.²⁴ These provisions do not themselves impose every performance guarantee found in an EPC schedule, but they make contractual conformity and intended function central to the statutory framework.

Article 95 adds an obligation to implement the contract according to its provisions and consistently with good faith, while also recognizing requirements arising from law, custom and the nature of the contract.²⁵ In a sophisticated EPC transaction, ‘the nature of the contract’ is important. The employer is not normally purchasing a pile of installed components; it is purchasing an integrated operating facility. That does not authorize a court or tribunal to invent output requirements absent from the bargain. It does, however, support an interpretation that gives meaningful effect to deliberately drafted performance guarantees rather than treating them as aspirational technical language.

The CTL also expressly permits incorporation of model documents or other documents by reference.²⁶ This matters in the typical EPC hierarchy, where General Conditions, Particular Conditions, Employer’s Requirements, technical specifications, schedules of guarantees, test codes, process design criteria and accepted contractor proposals may all form part of the same contract. The legal risk is not lack of documents but inconsistency among them. Article 104’s requirement that contractual terms be read consistently makes hierarchy clauses, document priority and express treatment of contradictions essential drafting tools.²⁷

The better approach is to place the legal status of performance guarantees beyond inference. Each guarantee should state whether it is (i) a minimum requirement and condition to acceptance, (ii) an economic guarantee subject to performance damages, or (iii) a design target with no independent remedial consequence. That taxonomy converts a technical schedule into an

²³ Civil Transactions Law, arts. 461–463.

²⁴ Id. art. 465.

²⁵ Id. art. 95.

²⁶ Civil Transactions Law, art. 46 (incorporation by reference to model documents or other documents).

²⁷ Civil Transactions Law, art. 104(2) (requiring interpretation by reference to mutual intent, custom, circumstances and consistency among contract terms).

intelligible allocation of legal risk. It also reduces the likelihood that a tribunal will have to reconstruct the parties' intention after the plant has failed a test and both sides have developed litigation positions.

Practitioner commentary on the CTL has rightly emphasized that Saudi construction contracts must now be read not only through the specific muqawala articles but also through the Code's general rules on good faith, damages, termination and agreed compensation.²⁸ Performance disputes are a prime example. They sit at the intersection of technical conformity and general law: the contractual test determines whether the metric was achieved, while the CTL determines what legal consequences can validly follow from failure.

Failure to Pass: Cure, Retesting, Taking Over, and Rejection

The Contractual Ladder of Remedies

A mature EPC contract should not jump directly from a failed performance test to termination. The usual architecture is a ladder. First, the test establishes a shortfall. Second, the contractor receives an opportunity to investigate, adjust, repair or modify the plant. Third, the test is repeated under defined conditions. Only after the available cure and retest process has been exhausted does the contract determine whether the plant must be rejected or may be accepted with a performance adjustment. The FIDIC Silver Book follows this general logic through its provisions on Tests on Completion, taking over, defects and Tests after Completion.²⁹

The CTL is compatible with that ladder. Article 466 provides that if the contractor breaches the contract during the work, the client may require compliance and correction within a reasonable period; if the breach is not corrected, the client may engage another contractor at the original contractor's expense or seek termination. Immediate termination is contemplated where the defect cannot be corrected or delay makes timely completion impossible.³⁰ Article 167 likewise addresses performance of an action at the debtor's expense where the debtor fails to perform and substitute performance is possible.³¹ These provisions favor cure where cure is meaningful, while preserving stronger remedies where correction is impossible or commercially futile.

That statutory structure should influence the drafting of performance-test remedies. A contract that labels a single failed test as an automatic, irreversible rejection event may be commercially appropriate for a genuinely fundamental criterion, but it should not be the default for every

²⁸ Watson Farley & Williams, *Saudi Arabia's Civil Transactions Law: Construction Contracts* (2023), discussing Articles 461–478 and the interaction of construction claims with the CTL's general principles, <https://www.wfw.com/articles/saudi-arabias-civil-transactions-law-construction-contracts/>.

²⁹ FIDIC Silver Book (2017), cls. 9–12 (testing, taking over, defects and tests after completion); William Godwin QC, *THE 2017 FIDIC CONTRACTS*, chs. 7–9 (Wiley 2020).

³⁰ Civil Transactions Law, art. 466.

³¹ *Id.* art. 167.

metric. Many performance defects are correctable through tuning, replacement of components, software changes, process optimization or limited redesign. The contract should therefore specify the number of retests, the time allowed for remedial work, responsibility for retest costs, the effect of operating the plant during that period and a long-stop date after which the employer can require a final election.

The legal significance of materiality also appears in Article 107. On breach of a bilateral contract, the other party may seek implementation or termination, with compensation where applicable, but the court may refuse termination if the unperformed part is insignificant compared with the obligation.³² This provision makes the minimum/economic distinction more than a drafting convenience. It provides a principled way to separate a performance failure that goes to the commercial substance of the project from a smaller shortfall for which termination would be disproportionate to the unperformed part.

The drafting consequence is straightforward. Minimum performance levels should be selected to represent the point at which continued acceptance is no longer a reasonable substitute for the promised bargain. If the employer wishes to preserve a termination or rejection right for failure below that line, the project record should show why the threshold is fundamental: financing assumptions, offtake obligations, regulatory permits, system-stability requirements, process integration, guaranteed product quality or another material dependency. A threshold selected merely as an aggressive negotiating position is more vulnerable to dispute than one tied to the project's actual commercial architecture.

Taking Over Without Full Performance

Taking over complicates the analysis because an employer may need the facility before every performance issue is resolved. In power and process projects, operation may be required to perform the very tests that determine final performance. Doosan Babcock illustrates the problem: the contractual scheme connected Tests on Completion, including performance tests, with taking over, but completion also depended on plant-wide conditions and performance by other participants.³³ The more integrated the facility, the less realistic it is to assume that testing occurs in a vacuum controlled solely by the EPC contractor.

The contract should therefore state expressly whether beneficial use, grid synchronization, introduction of feedstock, commercial dispatch or operation by the employer constitutes taking over, deemed taking over, temporary use, or no acceptance at all. It should also state which rights survive. Matthew Hall Ortech demonstrates why the distinction matters: process-plant conditions

³² Id. art. 107.

³³ Doosan Babcock Ltd v. Comercializadora de Equipos y Materiales Mabe Ltd [2013] EWHC 3201 (TCC), paras. 12–19 (tests on completion and the relationship between performance testing and taking over).

may allow operation and even taking over before all performance obligations are finally closed, while preserving defined performance remedies.³⁴

Saudi law adds another layer. Article 468 requires the client to take delivery when the contractor completes the work and makes it available, subject to legitimate reasons for refusal, while Article 469 links payment to taking delivery unless the parties agree otherwise.³⁵ In an EPC project, ‘completion’ for these purposes should be contractually defined with care. If the contract makes successful satisfaction of specified minimum performance tests part of completion, the employer has a stronger basis to contend that an asset below those minima has not yet been contractually completed. If the parties instead define taking over by physical availability alone, they should separately preserve the performance obligations and remedies that continue after that event.

The objective should not be to prevent the employer from operating the plant. It should be to prevent operation from creating unintended legal conclusions. A well-structured contract can permit temporary operation, revenue generation and testing while expressly reserving performance rights. That is particularly important where delay to commercial operation can itself create large losses or jeopardize project finance covenants. The law should not force an employer to choose between mitigating loss by operating the facility and preserving a legitimate performance claim.

Performance Liquidated Damages under the Civil Transactions Law

Validity, Harm, and the Evidentiary Starting Point

Performance liquidated damages are attractive because they convert a technical shortfall into a predictable financial consequence. Article 178 of the CTL now gives explicit statutory recognition to pre-agreed compensation: parties may determine the amount of compensation in advance, subject to the statutory exception concerning monetary obligations, and entitlement to such compensation does not require notice as a matter of default law.³⁶ In EPC drafting, this permits the parties to agree a formula for each unit of performance shortfall rather than attempting years later to prove the present value of lost production or increased operating cost.

Article 179 prevents that formula from becoming entirely self-executing. First, agreed compensation is not payable if the debtor proves that the creditor sustained no harm.³⁷ The structure is significant. A liquidated-damages clause does not disappear merely because the creditor does not initially prove actual loss. Saudi commentary and reported judicial practice indicate that the clause can establish an initial entitlement on breach, leaving the debtor to rebut

³⁴ Matthew Hall Ortech, [1997] EWHC Technology 352, paras. 47–51.

³⁵ Civil Transactions Law, arts. 468–469.

³⁶ Civil Transactions Law, art. 178.

³⁷ Id. art. 179(1).

the position by proving the absence of harm.³⁸ A reported Riyadh Commercial Court decision, Case No. 4530906759 of 1445H, has been cited for that reading of Article 179(1).³⁹

For performance damages, however, the ‘no harm’ issue is unusually technical. A plant that permanently produces less output will ordinarily impose an economic consequence, but the magnitude depends on utilization, market or offtake pricing, degradation, fuel costs, dispatch, curtailment, maintenance and financing assumptions. The contractor may argue that the employer suffered little or no loss because the plant was rarely dispatched at maximum output, because an offtaker did not require the missing capacity, or because another revenue mechanism insulated the employer. The employer may respond that the project was financed and priced on installed capability and that loss of capacity reduced the value of the asset even if every lost unit of output cannot be tied to a past sale.

The best defense to that dispute is not rhetorical. It is *ex ante* calibration. Performance damages should be derived from an economic model that identifies the loss reasonably associated with each unit of underperformance over the relevant period. The model need not be disclosed in full in the contract, but the negotiation record should preserve enough material to show that the formula was a genuine allocation of foreseeable economic risk rather than an arbitrary percentage. This is particularly important under a statute that expressly permits judicial review of agreed compensation.

Article 179 and the Partial-Performance Problem

The most important—and underappreciated—provision for performance damages is Article 179(2). It permits the court, on the debtor’s request, to reduce agreed compensation if the debtor establishes either that the amount is excessive or that the original obligation was partially performed.⁴⁰ The rule is mandatory; Article 179(4) invalidates agreements that contradict the Article.⁴¹ Practitioner analysis likewise emphasizes that parties cannot simply draft around the statutory power.⁴²

Performance disputes almost always involve partial performance. A contractor that delivers 1,470 MW against a 1,500 MW guarantee has performed 98 percent of the numerical result. If the contract imposes a fixed SAR 100 million sum for any failure to meet 1,500 MW, Article

38 Mahmoud Abdel-Baky, Rashed Z. Khalifah & Hamzeh Zu’bi, *Liquidated Damages Made Simple: What Saudi Arabia’s Civil Transactions Law Says*, GIBSON DUNN (15 May 2025), <https://www.gibsondunn.com/liquidated-damages-made-simple-what-saudi-arabia-civil-transactions-law-says/>.

39 Commercial Court in Riyadh, Case No. 4530906759 of 1445H, reported in Mahmoud Abdel-Baky, Rashed Z. Khalifah & Hamzeh Zu’bi, *Liquidated Damages Made Simple*, GIBSON DUNN (15 May 2025) (treating absence of harm as a matter for the debtor to establish under Article 179(1)).

40 Civil Transactions Law, art. 179(2).

41 Civil Transactions Law, art. 179(4).

42 Watson Farley & Williams, *Saudi Arabia’s Civil Transactions Law: Construction Contracts* (2023) (noting that Article 179 is mandatory and cannot be excluded by agreement).

179(2) invites an obvious argument: the obligation was substantially performed and the agreed compensation should be reduced. That argument becomes still stronger if the same fixed sum applies whether the shortfall is 1 MW or 100 MW.

A properly designed performance-damages formula responds to the statutory concern before the dispute arises. If damages are calculated per unit of shortfall—subject to appropriate caps, correction factors and thresholds—the formula already differentiates between 99.9 percent and 90 percent performance. In other words, proportionality is built into the contractual measure. The employer can then argue that Article 179’s concern with partial performance has already been reflected in the parties’ agreed valuation of the unperformed slice, rather than ignored by an all-or-nothing amount. This does not eliminate the court’s mandatory power, but it materially improves the clause’s explanatory and economic coherence.

The Advanced Polypropylene structure is instructive. The disclosed EPC arrangement allowed provisional acceptance when minimum levels were achieved but full guarantees were missed, provided the contractor paid liquidated damages at the prescribed rate for the shortfall; failure to meet the minimum levels could not be bought off with damages.⁴³ That design does two things at once. It treats partial performance above the minimum as commercially acceptable and prices it proportionally, while preserving the result obligation at the minimum threshold. It is a stronger fit with Articles 168 and 179 than a single undifferentiated liquidated sum.

The same logic explains why parties should distinguish a performance-damages cap from the minimum performance floor. A cap limits the contractor’s monetary exposure; it should not automatically transform an unacceptable plant into an acceptable one. If the damages cap is exhausted while the plant remains below the contractual minimum, the employer should retain the separate remedy assigned to failure of the minimum guarantee, subject to the CTL and the contract. Conversely, where the plant is above the minimum floor, the performance-damages cap can provide the contractor with a defined ceiling on the economic shortfall risk.

The commercial justification for liquidated damages remains powerful. In *Triple Point*, the United Kingdom Supreme Court emphasized that liquidated damages provide certainty, simplicity and a predictable allocation of risk where actual loss would otherwise be difficult and costly to quantify.⁴⁴ Those advantages are even more pronounced for performance shortfalls because long-term output loss can be highly sensitive to operating assumptions. Saudi law recognizes the same commercial need through Article 178, but Article 179 requires the drafting to remain connected to actual harm and partial performance.

⁴³ Advanced Polypropylene Company, Prospectus 47 (16 Nov. 2006), Capital Market Authority; compare *Energy Works (Hull) Ltd v. MW High Tech Projects UK Ltd* [2022] EWHC 3275 (TCC), para. 65.

⁴⁴ *Triple Point Technology, Inc v. PTT Public Co Ltd* [2021] UKSC 29, paras. 35, 74, 86 (describing the commercial function of liquidated damages as certainty, simplicity and risk allocation).

Excessiveness, Upward Adjustment, and the Contract Record

Article 179(2) also permits reduction where the agreed compensation is excessive.⁴⁵ The statutory text does not convert every difference between agreed and later-proved loss into a remeasurement exercise. Reported Saudi commentary suggests that the power is directed at excessiveness rather than ordinary divergence.⁴⁶ Still, the performance-damages clause is more defensible when its rate can be explained by reference to the economic characteristics known at contract formation.

The opposite possibility must also be remembered. Article 179(3) permits the creditor to seek an increase where fraud or gross negligence by the debtor caused harm exceeding the agreed amount.⁴⁷ A performance-damages cap should therefore not be analyzed in isolation from Article 173 and the contract's fraud or gross-negligence carve-outs. If the contractor has deliberately manipulated test data, concealed a material design defect or engaged in conduct amounting to gross negligence, the ordinary capped performance remedy may not be the end of the analysis.

Where compensation has not been contractually fixed, Article 180 directs the court to the general compensation rules and limits ordinary contractual recovery, absent fraud or gross negligence, to harm that could have been anticipated at the time of contracting.⁴⁸ Articles 136 and 137 provide for full compensation and recognize actual loss and lost profit where legally established.⁴⁹ In performance disputes, these provisions reinforce the value of contemporaneous economic modeling: even where a liquidated formula does not apply, the project documents should show why a particular output or efficiency shortfall was economically foreseeable.

The Relationship Between Performance Damages and Other Remedies

Exclusive Remedy Clauses and Remedial Sequencing

An EPC contract rarely contains only one remedy. The employer may have rights to require repair, retest, recover performance damages, reject the plant, terminate, call security, recover general damages, or pursue indemnities. If the relationship among these rights is not expressed, performance disputes can become arguments about double recovery, election, exclusivity and whether payment of performance damages purchases the contractor's release from further responsibility.

⁴⁵ Civil Transactions Law, art. 179(2); Abdel-Baky, Khalifah & Zu'bi, Liquidated Damages Made Simple, GIBSON DUNN (15 May 2025) (recommending that drafting expressly address partial performance).

⁴⁶ Mahmoud Abdel-Baky, Rashed Z. Khalifah & Hamzeh Zu'bi, Liquidated Damages Made Simple: What Saudi Arabia's Civil Transactions Law Says, GIBSON DUNN (15 May 2025), <https://www.gibsondunn.com/liquidated-damages-made-simple-what-saudi-arabia-civil-transactions-law-says/>.

⁴⁷ Civil Transactions Law, art. 179(3).

⁴⁸ Id. art. 180.

⁴⁹ Id. arts. 136–137.

The cleanest structure is sequential. During the cure period, the contractor remains obliged to correct the shortfall and retest. If the plant ultimately satisfies the guarantee, no permanent performance damages should arise, although delay or retest costs may be separately addressed. If the plant satisfies the minimum level but remains below the economic guarantee at the final test, the employer may accept it and recover the specified performance damages. If the plant fails the minimum level at the long-stop, the contract should identify the employer's election: further cure, negotiated acceptance, price reduction, rejection or termination. The parties should state whether a negotiated acceptance below the minimum converts the shortfall into an agreed economic remedy or preserves other claims.

The FIDIC Silver Book contains a version of this remedial philosophy in its Tests after Completion regime where specified performance damages may operate as the contractual response to a quantified failure.⁵⁰ Process-plant jurisprudence likewise recognizes the commercial logic of accepting reduced performance together with liquidated damages.⁵¹ The value of the structure is not its label but its finality: the parties know when an underperforming but usable asset is accepted and what price is attached to that acceptance.

Saudi law permits significant contractual allocation of liability. Article 173 allows agreement to exempt a debtor from compensation for contractual non-performance or delay except where fraud or gross negligence is involved, while liability for a harmful act cannot be excluded.⁵² Contemporary analysis of the CTL reads this as permitting contractual limitation and exclusion of contractual liability subject to those statutory boundaries.⁵³ Accordingly, a clause stating that properly calculated performance damages are the employer's sole monetary remedy for a defined economic shortfall is in principle capable of meaningful effect, but it must be drafted consistently with the mandatory rules of Article 179 and the carve-outs required by Article 173.

The distinction between sole monetary remedy and sole remedy altogether is important. An employer may agree that performance damages are the exclusive financial compensation for a shortfall above the minimum level while still preserving the contractor's duty to complete a defined cure and testing process. Conversely, where the contract states that payment of performance damages deems the relevant performance test passed, it should be clear that this consequence applies only to the economic guarantee and not to safety, legality, product quality, emissions or another minimum requirement that cannot commercially be waived by payment.

⁵⁰ FIDIC Silver Book (2017), cl. 12.4 (remedies for failure to pass tests after completion where performance damages are specified).

⁵¹ Matthew Hall Ortech, [1997] EWHC Technology 352, para. 51 (reduced performance may be accepted together with liquidated damages under a properly structured plant contract).

⁵² Id. art. 173.

⁵³ Nabeel Ikram & Giuliano Parlascino, Saudi Arabia's Civil Transactions Law, INTERNATIONAL BAR ASSOCIATION (25 July 2024) (Article 173 permits contractual limitation or exclusion of contractual compensation, subject to fraud or gross negligence, while tort liability cannot be excluded).

Termination and the Material Shortfall

Termination should be reserved for the category of failure that defeats the project bargain. Article 107's recognition that termination may be refused where the unperformed part is insignificant gives statutory support to a materiality analysis.⁵⁴ Article 466 separately permits immediate termination for an uncorrectable defect or delay making timely completion impossible.⁵⁵ Read with Article 168, these provisions suggest a coherent hierarchy: failure to achieve a specified result establishes non-performance; the seriousness and correctability of that non-performance help determine the appropriate remedy.

The minimum performance threshold should therefore do legal work. It should not merely duplicate the guaranteed target at a slightly lower figure. It should identify the point at which the employer's acceptance of money is no longer an adequate substitute for the promised asset. For a power project, that may relate to minimum net dependable capacity, stable operation, heat rate, emissions compliance or grid-code performance. For a process facility, it may relate to throughput, product specification, recovery, feedstock envelope or a safety-critical operating parameter. The threshold should be integrated with financing, offtake and regulatory requirements rather than negotiated in isolation by the technical and legal teams.

Where the employer elects termination, the contract should also address the relationship between performance damages accrued before termination and general damages thereafter. The general law of liquidated damages in other jurisdictions often distinguishes pre-termination accrual from post-termination loss; Triple Point is a leading example.⁵⁶ Saudi law will ultimately be governed by the CTL and the parties' wording, but the comparative lesson is useful: the contract should say what survives termination rather than leave a tribunal to infer the intended boundary.

Testing Integrity, Causation, and Employer Conduct

A Performance Test Is an Evidentiary Event

A performance guarantee is only as reliable as the test that measures it. Test procedures should therefore be treated as evidentiary architecture, not technical annexes delegated entirely to engineers. They determine whether a legal obligation of result has been achieved. The procedure should define instrumentation, calibration, duration, stable operating conditions, permitted outages, ambient corrections, feedstock or fuel characteristics, degradation corrections, measurement uncertainty, test tolerances, data validation, independent witnessing, treatment of abnormal conditions and the hierarchy between international test codes and project-specific rules.

⁵⁴ Id. art. 107.

⁵⁵ Civil Transactions Law, art. 466.

⁵⁶ Triple Point, [2021] UKSC 29, paras. 74–86.

The classic process-plant literature recognizes this point. Matthew Hall Ortech noted that performance tests may occur under conditions different from the basic design case and that correction methods are therefore needed to translate observed results to the guaranteed conditions.⁵⁷ The proposition is simple but legally important. Without agreed correction methodology, the parties may dispute not only the measured result but what the result would have been under the contractual reference conditions. At that stage, an output guarantee can become an expert-evidence dispute about thermodynamics rather than a clean contractual metric.

The testing schedule should also allocate control. Who operates the plant? Who supplies fuel or feedstock? Who controls grid dispatch? Who determines when the plant is stable enough to begin the test? Who can abort a run? What happens if the employer's upstream or downstream facility is unavailable? Doosan Babcock demonstrates that performance testing can depend on other project participants and on the wider plant being sufficiently complete.⁵⁸ A contractor should not be deemed to have failed a result that the employer's interface conditions made impossible to demonstrate, while an employer should not lose its guarantee because the contractor can repeatedly invoke minor test irregularities.

Causation, Contributory Fault, and Good Faith

The CTL provides a direct answer to employer contribution. Article 172 applies Article 128 where the creditor, by its own fault, contributes to or aggravates the harm arising from non-performance or delay; Article 128 permits compensation to be wholly or partially forfeited in proportion to that contribution.⁵⁹ In performance testing, this makes interface records critical. If the employer supplied off-specification fuel, imposed abnormal dispatch instructions, failed to provide utilities, or prevented the contractor from conducting a valid retest, the causal effect of that conduct may reduce the recoverable compensation even where a technical shortfall exists.

The principle should not be confused with automatic excuse. The contractor may have assumed defined input ranges or interface risks. A Silver Book contractor commonly accepts substantial responsibility for integration and end performance. The first question is therefore contractual allocation; causation comes next. If the contract states that the guarantee applies across a particular fuel range, operation within that range cannot later be characterized as employer fault merely because it made the test more difficult. The technical schedule must define the boundary between transferred performance risk and employer-controlled test conditions.

Good faith reinforces this discipline. Article 95 requires implementation consistent with good faith.⁶⁰ Article 29 prohibits abusive exercise of rights, including exercise solely to harm another,

⁵⁷ Matthew Hall Ortech, [1997] EWHC Technology 352, para. 50 (noting the need for performance-test corrections when test conditions differ from the design case).

⁵⁸ Doosan Babcock, [2013] EWHC 3201 (TCC), paras. 12–19.

⁵⁹ Id. arts. 128, 172.

⁶⁰ Id. art. 95.

grossly disproportionate exercise, or exercise for an unlawful or unintended purpose.⁶¹ These provisions should not be used to rewrite a clear performance guarantee or excuse a contractor from a test it failed. They are more appropriately directed at conduct surrounding administration: refusing a valid test window for tactical reasons, manipulating dispatch to cause failure, withholding data needed for corrections, or insisting on a formal retest that has become objectively impossible because of the employer's own conduct.

For the same reason, contemporaneous test governance is a dispute-avoidance tool. Daily logs, calibration certificates, raw data, operator instructions, fuel or feedstock samples, weather records, control-system historian data, notices of abnormal conditions and agreed test calculations should be preserved as contract records. A performance dispute that reaches arbitration years later will often turn less on elegant legal doctrine than on whether the parties can reconstruct precisely what the plant was doing during a four-hour or seventy-two-hour test.

Comparative Lessons from International EPC Case Law

Result Obligations and Technical Standards: MT Højgaard

Comparative case law is useful not because English decisions govern Saudi contracts, but because mature engineering disputes expose recurring drafting failures. MT Højgaard is the clearest example of the tension between compliance with a technical standard and responsibility for a promised result. The contractor designed offshore wind foundations in accordance with an industry standard, but the incorporated technical requirements imposed a more demanding design-life obligation. The Supreme Court held the contractor responsible under the contractual requirement.⁶²

For Saudi drafting, the lesson is not that every performance statement should override every design standard. The lesson is that the contract must define the relationship between them. Article 104 requires the agreement to be read as a coherent whole, while Article 168 makes the presence of a specific result legally consequential.⁶³ If the Employer's Requirements say both 'design in accordance with Standard X' and 'guarantee Result Y,' the contract should state whether compliance with Standard X is merely a minimum method requirement or a safe harbor that satisfies the result obligation. Silence can transfer a risk that neither technical team priced consciously.

61 Id. art. 29.

62 MT Højgaard, [2017] UKSC 59, paras. 31–34, 51–54.

63 Civil Transactions Law, arts. 95, 104, 168, 179.

Acceptance Architecture: Matthew Hall Ortech

Matthew Hall Ortech remains valuable because it treats plant delivery as a sequence rather than a single completion event. The judgment describes construction completion, taking over, performance testing, acceptance, defect correction and performance guarantees as related but distinct stages.⁶⁴ It also recognizes the commercial possibility that the purchaser may accept a reduced performance level with liquidated damages.⁶⁵

The lesson for Saudi EPC contracts is to make certificates and legal consequences correspond. If a Taking-Over Certificate transfers care but does not settle economic performance, say so. If a Performance Acceptance Certificate closes the contractor's obligation to improve an economic guarantee, define that finality. If minimum performance remains unachieved, avoid issuing a certificate whose wording can later be characterized as unconditional acceptance. The certificate should record reservations and the surviving obligations with the same care as the underlying clause.

Absolute and Economic Guarantees: Energy Works (Hull)

Energy Works (Hull) demonstrates the value of separating different categories of performance promise. The EPC contract listed absolute performance guarantees and economic performance guarantees; failure of an absolute guarantee could justify rejection, while the economic guarantees addressed a different level of underperformance.⁶⁶ The distinction is closely aligned with the model proposed here.

It also helps resolve a drafting question that Article 179 makes pressing: what does 'partial performance' mean when a guarantee is numerical? For an economic guarantee, partial performance can be reflected in a sliding formula. For an absolute guarantee, the issue is different. The contractor may have delivered many aspects of the plant, but if a single threshold defines commercial viability—safe operation, legal emissions, mandatory product quality, stable minimum load—the fact that other obligations were performed does not convert the failed threshold into a minor economic shortfall. The contract should explain that role rather than assume the label 'absolute' will do all the work.

Solar and Integrated-Plant Disputes

Other cases demonstrate the breadth of the issue. Toucan Energy Holdings v. Wirsol Energy arose from EPC arrangements for multiple solar projects and illustrates how performance

⁶⁴ Matthew Hall Ortech Ltd v. Tarmac Roadstone Ltd [1997] EWHC Technology 352, paras. 19–21, 47, 50–51 (discussing plant testing, performance guarantees, acceptance and reduced performance coupled with liquidated damages).

⁶⁵ Matthew Hall Ortech, [1997] EWHC Technology 352, para. 51 (reduced performance may be accepted together with liquidated damages under a properly structured plant contract).

⁶⁶ Energy Works (Hull), [2022] EWHC 3275 (TCC), paras. 63–65.

obligations, defects, acceptance and termination can interact across an operating portfolio.⁶⁷ Doosan Babcock shows that a contractor's ability to complete performance tests may depend on installation, commissioning and wider-plant conditions beyond the supplied equipment.⁶⁸ Neither case supplies a Saudi rule. Both warn against treating performance testing as an isolated binary event divorced from interfaces, operation and the contractual sequence.

The comparative cases also confirm a broader drafting principle: technical remedies must be designed before the project begins, not invented after failure. Once the asset is operating, the parties face sunk capital, financing pressure, offtake obligations and operational urgency. At that point, rejection may be economically irrational even if legally available. A contract that provides a credible path from failed test to cure, retest, quantified economic acceptance or fundamental rejection gives the parties a dispute-avoidance mechanism when their leverage and incentives are otherwise most polarized.

Drafting and Project-Counsel Recommendations for Saudi EPC Projects

Separate the Three Categories of Technical Commitment

Saudi EPC schedules should classify technical commitments into three categories. First, design criteria and assumptions identify the basis on which the contractor is to engineer the works; they should not automatically become guaranteed outcomes. Second, economic performance guarantees identify outputs or consumptions for which a shortfall can be accepted at an agreed monetary adjustment. Third, minimum or absolute performance requirements identify outcomes below which the employer is not obliged to accept the facility. This taxonomy should appear in the contract itself, not merely in internal engineering documents.

The distinction is supported by actual Saudi project structures. Ma'aden's public disclosures describe EPC contracts with performance liquidated damages where guaranteed performance is not achieved.⁶⁹ The Advanced Polypropylene prospectus describes a still more explicit separation between performance guarantees and minimum performance levels, with liquidated damages available for shortfalls above the minimum but insufficient to excuse failure below it.⁷⁰ These examples show that performance risk can be divided with considerably more precision than a single generic fitness-for-purpose clause.

67 Toucan Energy Holdings Ltd v. Wirsol Energy Ltd [2021] EWHC 895 (Comm) (dispute arising from EPC arrangements for solar projects, including contractual performance obligations and termination consequences).

68 Doosan Babcock Ltd v. Comercializadora de Equipos y Materiales Mabe Ltd [2013] EWHC 3201 (TCC), paras. 12–19 (tests on completion and the relationship between performance testing and taking over).

69 Saudi Arabian Mining Company (Ma'aden), Rights Issue Prospectus 213, 221, 227 (2014), Capital Market Authority (performance damages and liability structures in Saudi industrial EPC contracting).

70 Advanced Polypropylene Company, Prospectus 47 (16 Nov. 2006), Capital Market Authority (performance shortfall damages capped at ten percent of the contract price while minimum performance levels remained mandatory).

Draft Performance Damages for Article 179, Not Against It

The performance-damages formula should be drafted on the assumption that Article 179 will apply. The clause should identify the triggering metric, the contractual reference conditions, the unit of shortfall, the rate per unit, the maximum amount, the treatment of multiple simultaneous shortfalls and whether different damages are cumulative or subject to an aggregate cap. Most importantly, the formula should scale with the degree of underperformance wherever the economic loss itself scales with performance.⁷¹

The project team should preserve the economic rationale for the rate. That may include financial-model sensitivities, expected utilization, offtake pricing, fuel-cost curves, degradation assumptions, operating life and the cost of substitute capacity. The objective is not to prove actual damages in advance but to demonstrate that the agreed formula was a commercially serious estimate or allocation of a foreseeable risk. This record is particularly valuable if the contractor later seeks reduction for excessiveness or partial performance under Article 179.

The clause should also avoid applying performance damages to a monetary payment obligation, given Article 178's statutory limitation and Saudi public-policy concerns concerning compensation for payment delay. Performance damages should attach to failure of the non-monetary technical obligation—the promised output, efficiency or other result—not to late payment of money.

Define the Cure, Retest, and Long-Stop Process

The contract should establish a complete testing ladder: initial test; notice of failure; contractor investigation; corrective work; retest; permitted number of attempts; allocation of retest costs; long-stop date; and employer election. FIDIC provides a useful structural starting point through its testing, taking-over and defects clauses, but Particular Conditions and technical schedules must be tailored to the technology and project finance structure.⁷²

The long-stop mechanism should answer the commercial question that often remains hidden until late in commissioning: how long must the employer tolerate repeated attempts to reach performance? The answer may be a fixed period, a number of retests, or a combination. After the long-stop, the employer should be able to choose among specified outcomes, subject to the nature and materiality of the shortfall and the CTL's remedial framework.⁷³

⁷¹ Civil Transactions Law, art. 179; Commercial Court in Riyadh, Case No. 4530906759 of 1445H, reported in Abdel-Baky, Khalifah & Zu'bi, GIBSON DUNN (15 May 2025).

⁷² FIDIC Silver Book (2017), cls. 9, 10, 11 & 12; FIDIC, THE FIDIC SUITE OF CONTRACTS.

⁷³ Civil Transactions Law, arts. 107, 167, 168, 466.

Protect Testing Integrity and Interface Responsibility

Every guaranteed metric should have a corresponding test procedure capable of producing legally reliable evidence. The procedure should include correction curves for ambient or input conditions, rules for measurement uncertainty, calibration standards, test duration, stable-operation criteria and a process for resolving disagreement over raw data. The parties should determine whether an independent engineer or specialist test agency has a role and whether its determination is final, provisional, or subject to dispute resolution.

Interface matrices are equally important. Employer-supplied fuel, feedstock, utilities, grid availability, water quality, upstream and downstream facilities, permits and operator actions should be assigned clearly. The contract should distinguish a contractor performance failure from a failed test caused by employer conditions. That allocation should then be aligned with Articles 128 and 172 on creditor contribution and with the general duties of good faith and non-abusive exercise of rights.⁷⁴

Integrate Acceptance with Financing and Offtake

Performance guarantees should be tested against the project's revenue contracts, not drafted solely from the EPC contractor's technical proposal. A power purchase agreement may require a minimum dependable capacity or impose availability consequences. A feedstock or offtake arrangement may assume a particular throughput or product specification. Financing documents may condition completion on technical tests or independent-engineer certification. The EPC minimum performance floor should be set with those dependencies in view.

The Saudi project disclosures examined above illustrate the point. Advanced Polypropylene's disclosed EPC acceptance regime sat alongside offtake arrangements structured around nameplate capacity and revenue stability.⁷⁵ Ma'aden's EPC structures transfer substantial design and construction risk to contractors and use performance damages as part of that allocation.⁷⁶ Performance remedies are therefore not isolated construction-law devices. They are mechanisms for protecting the project's bankability and revenue model.

Preserve the Distinction Between a Price for Shortfall and a License to Fail

The most important drafting principle is conceptual. Performance damages should purchase an agreed degree of economic underperformance; they should not purchase the right to deliver an asset below the project's minimum functional identity. The contract should say so expressly.

⁷⁴ Id. arts. 29, 95, 128, 172.

⁷⁵ Advanced Polypropylene Company, Prospectus 47–48 (16 Nov. 2006), Capital Market Authority (linking plant acceptance, performance shortfall remedies and downstream offtake economics).

⁷⁶ Saudi Arabian Mining Company (Ma'aden), Rights Issue Prospectus 213 (2014), Capital Market Authority (contractor assumes major design and construction risks; performance liquidated damages may follow failure to achieve guaranteed performance).

Payment of damages for an economic guarantee may deem that guarantee satisfied for acceptance purposes. Payment should not deem a failed minimum requirement satisfied unless the employer makes a separate, informed election to accept that result.

This distinction aligns the contract with both Article 168 and Article 179. Article 168 supports enforcement of a deliberately promised result. Article 179 ensures that monetary compensation for a shortfall remains connected to harm, proportionality and partial performance.⁷⁷ The two provisions are not in tension. Together they encourage a more sophisticated bargain: strict performance where the result is indispensable, and calibrated compensation where the result is economically substitutable.

Conclusion

The legal importance of performance testing in Saudi EPC projects will grow as the Kingdom's project portfolio becomes more operationally complex. In output-based assets, the employer's bargain is not exhausted by physical construction. The economic product is a functioning facility with specified capabilities. The CTL now provides a clear doctrinal basis for recognizing that reality.

Article 168 is the central provision. It makes the distinction between reasonable-care obligations and result obligations explicit and provides that a required specific result is performed only when achieved. Properly drafted EPC output guarantees can therefore operate as genuine obligations of result under Saudi law. Articles 461, 463 and 465 reinforce the importance of contractual conformity and intended purpose, while Articles 95 and 104 require performance and interpretation to respect the contract as an integrated commercial instrument.

The remedial analysis is more nuanced. Articles 167, 466 and 107 support cure, substitute performance and termination according to the nature and materiality of the breach. Articles 128 and 172 prevent an employer from recovering compensation for harm materially caused or aggravated by its own conduct. Articles 178 to 180 recognize pre-agreed compensation but subject it to statutory safeguards. Article 179 is particularly significant because performance shortfalls are, by definition, frequently cases of partial performance.

For that reason, the strongest Saudi EPC architecture is not an all-or-nothing performance clause. It is a two-tier system. Minimum or absolute performance requirements define the floor below which the employer has not received an acceptable version of the promised asset. Economic performance guarantees define the zone in which the employer will accept the plant with a calibrated monetary adjustment. Comparative engineering cases and publicly disclosed Saudi project structures show that this distinction is commercially mature.⁷⁸ The CTL gives it a coherent statutory foundation.

⁷⁷ Civil Transactions Law, arts. 168, 178–180.

The practical consequence for project counsel is equally clear. Performance schedules must be drafted with the same legal care as limitation-of-liability and termination clauses. Test procedures should be treated as evidentiary instruments. Performance-damages formulas should be economically modeled and proportionate to shortfall. Minimum thresholds should be tied to real project dependencies. Acceptance certificates should state what obligations survive. Interface responsibility should be documented before testing begins. If these matters are addressed early, the contract can turn a failed test from a crisis into a managed allocation of risk.

Saudi law does not require parties to choose between strict EPC performance and commercial flexibility. It permits both. The more difficult task is to identify, before the plant is built, which promises must be kept as results and which shortfalls can be priced. That is where the legal architecture of the EPC contract does its most valuable work.

78 Energy Works (Hull) Ltd v. MW High Tech Projects UK Ltd [2022] EWHC 3275 (TCC), para. 65; Advanced Polypropylene Company, Prospectus 47 (16 Nov. 2006), Capital Market Authority.

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